
CrowPanel 2.4-inch ESP32 HMI

Technical Specifications & Hardware Reference

Repository: crowpanel-2.4inch-esp32-hmi

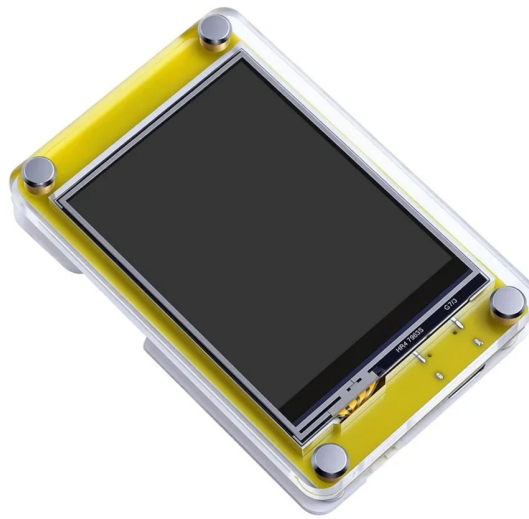


Figure 1 — Front view of the CrowPanel 2.4-inch ESP32 HMI.

A compact ESP32-based human-machine interface platform integrating a 320 × 240 TFT display, resistive touchscreen, wireless connectivity, TF card storage, UART, I²C, GPIO, audio and battery interfaces.

Document scope: Hardware architecture, electrical interfaces, display/touch specifications, connector mapping, dimensions, software support and typical embedded application use cases.

1. Hardware Overview

The CrowPanel 2.4-inch ESP32 HMI is an integrated embedded development module built around the ESP32-WROOM-32-N4. It provides a compact graphical HMI with local touch input and multiple expansion interfaces for connected embedded systems.

Front View

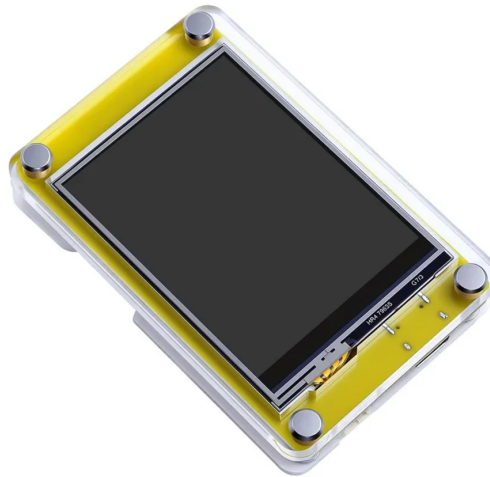


Figure 2 — Front view showing the display and integrated controller board.

Rear / Interface View

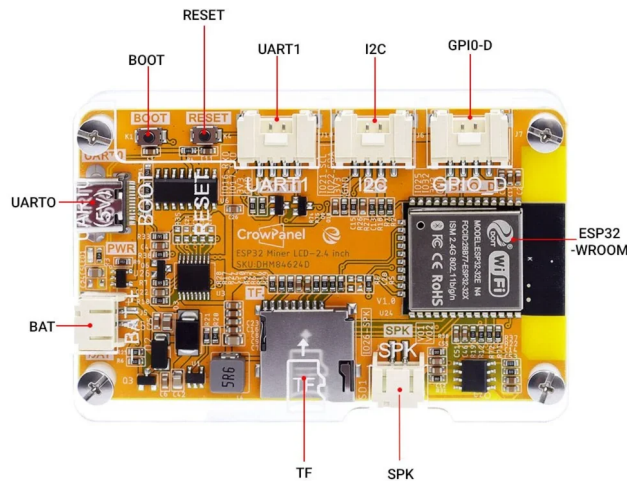


Figure 3 — Rear/component view identifying connectors, buttons and major interfaces.

2. Mechanical & PCB Reference

Mechanical Dimensions

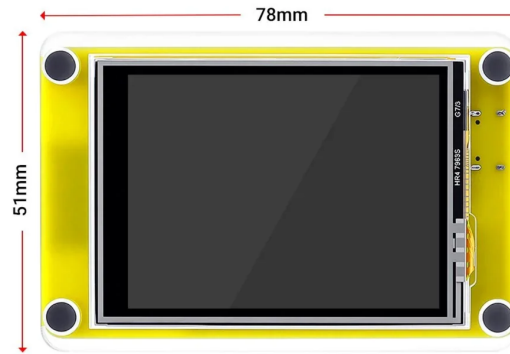


Figure 4 — Overall module reference dimensions: approximately 78 mm × 51 mm.

PCB Layout

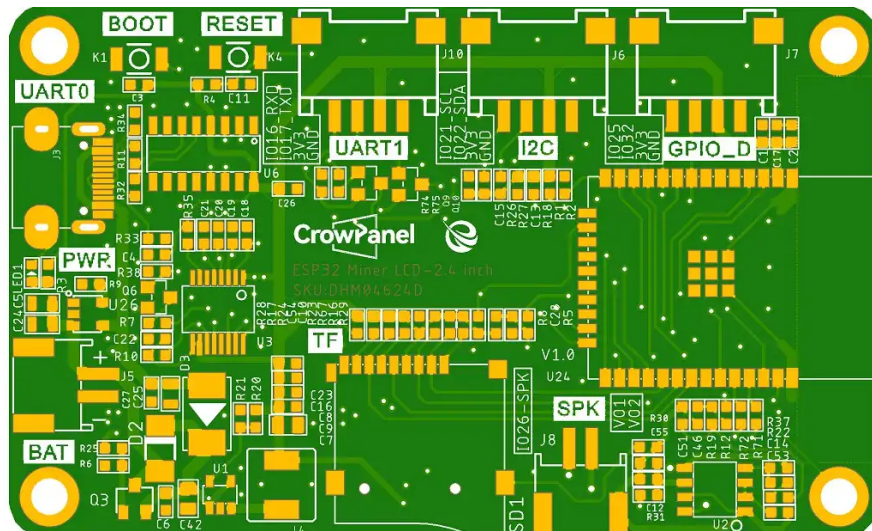


Figure 5 — PCB layout reference showing component placement and interface locations.

Mechanical note: The 78 × 51 mm dimension is the overall board/module reference shown in the supplied manufacturer imagery. Allow additional clearance for connectors, cables, mounting hardware and enclosure walls during mechanical design.

3. Technical Specifications

Parameter	Specification
Microcontroller	ESP32-WROOM-32-N4
CPU	Dual-core 32-bit Xtensa LX6
Maximum CPU Frequency	Up to 240 MHz
Flash	4 MB
Display	2.4-inch TFT LCD
Display Resolution	320 × 240 pixels
Display Controller	ILI9341V
Touch Interface	Resistive touchscreen
Wireless	2.4 GHz Wi-Fi 802.11 b/g/n; Bluetooth
External Storage	TF / microSD card
Audio	Speaker interface with onboard amplifier circuit
Battery Input	3.7–4.2 V lithium battery interface with charging circuit
USB / Serial	USB / USB-C interface for UART0, power and serial communication
UART	UART1 via 4-pin connector
I ² C	Dedicated 4-pin connector
GPIO	GPIO-D 4-pin connector; two user GPIO lines
Dimensions	Approx. 78 mm × 51 mm

Note: Specifications are consolidated from the supplied board/product information and reference images. Connector labels and GPIO assignments should be verified against the board revision/schematic before use in production hardware.

4. Interfaces & Connector Mapping

Interface	Signals / GPIO	Connector / Function
UART0	USB serial interface	USB / USB-C; programming, power and serial logging
UART1	RX = GPIO16 TX = GPIO17	4-pin UART1 connector
I ² C	SDA = GPIO22 SCL = GPIO21	4-pin I ² C connector
GPIO-D	GPIO25 GPIO32	4-pin digital I/O connector
Speaker	GPIO26	SPK connector; onboard amplifier
TF / microSD	MOSI = GPIO23 MISO = GPIO19 SCLK = GPIO18 CS = GPIO5	Onboard TF card slot; SPI
Battery	3.7–4.2 V	BAT connector; Li-ion battery interface
BOOT	Boot control	Onboard push button
RESET	Reset control	Onboard push button

5. Display Interface

Signal	ESP32 GPIO	Function
TFT MISO	GPIO12	SPI data input
TFT MOSI	GPIO13	SPI data output
TFT SCLK	GPIO14	SPI clock
TFT CS	GPIO15	Display chip select
TFT DC	GPIO2	Data / command control
TFT RESET	-1	Reset handled by board / software configuration
TFT Backlight	GPIO27	Backlight control

6. Resistive Touch Interface

Signal	ESP32 GPIO	Function
Touch CS	GPIO33	Touch controller chip select
Touch IRQ	GPIO36	Touch interrupt / input
Touch SCLK	GPIO14	Shared SPI clock
Touch MOSI	GPIO13	Shared SPI data output
Touch MISO	GPIO12	Shared SPI data input

The touchscreen shares the SPI bus with the TFT display. Software must manage the individual chip-select signals when accessing the display and touch controller.

7. TF / microSD Interface

Signal	GPIO	Function
MOSI	GPIO23	SPI data output
MISO	GPIO19	SPI data input
SCLK	GPIO18	SPI clock
CS	GPIO5	TF card chip select

The TF card can be used for image storage, configuration files, data logging, HMI assets, fonts and application data.

8. Software & Development Support

Platform / Framework	Use
Arduino IDE	Rapid firmware development and hardware examples
PlatformIO	Project-based embedded development and dependency management
ESP-IDF	Espressif native development framework
MicroPython	Interactive Python-based embedded development
LVGL	Advanced graphical user interfaces and HMI development
ESPHome	IoT device integration and automation
Home Assistant	Smart-home / IoT integration

9. Recommended Libraries

Library / Component	Typical Use
Adafruit GFX	Graphics primitives and text rendering
Adafruit ILI9341	TFT display driver
XPT2046_Touchscreen	Resistive touchscreen controller interface
SPI	Display, touch and storage communication
SD	TF / microSD card access
Wire	I ² C communication
WiFi	ESP32 Wi-Fi connectivity
Bluetooth	ESP32 Bluetooth connectivity
LVGL	Advanced embedded GUI development

10. Typical Applications

Automotive HMI and vehicle status displays • Battery and BMS monitoring interfaces • CAN bus visualization and diagnostic displays • Railway / train HMI prototypes • Industrial automation panels • IoT sensor dashboards • Environmental monitoring systems • Robotics and embedded control interfaces • Smart-home control panels • Data logging and portable embedded instruments • Educational and rapid-prototyping platforms

11. Engineering Notes

This document is intended as a technical reference for development and prototyping. GPIO availability can be affected by onboard peripherals, boot configuration, shared buses and board revisions. Verify electrical levels, connector pin order, current limits and peripheral conflicts against the applicable schematic before connecting external hardware.

The battery interface is intended for a 3.7–4.2 V lithium battery according to the supplied product information. Use an appropriate protected cell and observe the charging/power requirements of the specific board revision.

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